

Measure Theory, Probability and Statistics

CM1 — Course presentation and complete Chapter 1

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- 1 Course presentation and rules
- 2 Chapter 1: Around Lebesgue outer measure
 - Reminder on sets
 - Lebesgue outer measure
 - Carathéodory criterion
 - The Vitali set

Why this course?

Mathematical objective

Build a rigorous framework for discussing probability, statistics, expectation, integration, convergence and limit theorems.

Core idea

A probability is a measure with total mass 1:

$\text{sigma-algebra} \longrightarrow \text{measure} \longrightarrow \text{integral} \longrightarrow \text{probability}$
 $\longrightarrow \text{statistics (in S8)}.$

General progression of the semester

Part I: measure theory

- 1 Lebesgue outer measure
- 2 Sigma-algebras and Borel sigma-algebra
- 3 Measures
- 4 Measurable functions
- 5 Lebesgue integral
- 6 Multiple integrals
- 7 Decomposition of measures
- 8 L^p spaces

Part II: probability theory

- 9 Real random variables
- 10 Random vectors
- 11 Conditioning
- 12 Gaussian vectors
- 13 Convergences and limit theorems

Organisation of tutorials

Work rhythm

In principle, each week's tutorial is based on the content covered in the previous lecture. If the course falls behind schedule, the tutorial will be adjusted to the content actually covered.

Group rotation

The tutorials will run with rotations between the groups supervised by Benoit Duvocelle, Professor Eric and Professor Tina.

Objective in tutorials

Learn to write complete solutions, explicitly citing the theorems used.

Structure of tutorial sheets

- ① **direct applications of the course**: at most ten basic exercises.
- ② **For practice**: more technical exercises, including Kaoyan exercises.
- ③ **Going further**: proofs omitted from lectures and theoretical extensions.

Role of tutorial teachers

Common principle

Each group must receive the same minimum content, the same level of rigor and the same working rules.

Coordination

After each tutorial, the main difficulties and the progress of the groups will be reported. At the end of the tutorial, the students will upload their handwritten notes to a shared drive in order to identify frequent mistakes.

Languages and documents

Course materials

All course documents will be provided in French and in English.

Working languages

Depending on the groups and situations, explanations may use French, Chinese or English.

Mathematical reference

The notation, definitions and the reference results are those in the course notes and the slides.

Three marks

$$\text{Final mark} = 0.4 \times \text{Exam 1} + 0.5 \times \text{Exam 2} + 0.1 \times \text{Participation.}$$

Participation

Participation is graded out of 10:

- an important part comes from attendance in lectures and tutorials;
- active participation and real work count;
- lateness, absences, prohibited use of devices and use of AI may lower the mark.

One week before each exam

A detailed syllabus will be given to specify exactly the required topics.

What may be asked

- one proof from a given list;
- definitions or theorems to be able to state precisely;
- exercises close to the tutorials;
- an exercise “somewhat” difficult to distinguish the best papers.

Rules in tutorials: devices and AI

Principle

The goal of the tutorial is to learn to reason by yourself. AI is therefore forbidden during tutorials.

Devices

- phones: forbidden;
- computers: forbidden;
- tablets: allowed only to consult the tutorial sheet.

At the desk/tablet rule

No AI, no phone, no computer. The tablet may only be used to consult the worksheet.

Attendance and communication

Attendance

Attendance in lectures and tutorials is important and contributes to the participation mark. An attendance check may be organized with verification of the number of students present.

Communication

All course documents will be placed on Dingding.

Ungraded feedback

At the end of each tutorial, send your work on Dingding in a dedicated space. This will be processed by an AI, which may summarize the points on which students are struggling.

What I expect from you

- 1 Learn the course
- 2 Know how to move from one language to another: sets, measure, probability.
- 3 Write short but complete proofs.
- 4 Compute cleanly: sums, integrals, distributions, expectations.
- 5 Ask questions as soon as a word or notation is blocking.
- 6 Active participation in tutorials and curiosity.

Objectives from Chapter 1

- Understand the *problem of measure* and the Lebesgue solution.
- Know how to handle the outer measure and the Carathéodory criterion.
- Understand, through the example of Vitali, why we cannot measure all the subsets of $\mathbb{R}$.
- Handle the indicator functions and the logical operations: union, intersection, complement.

Introduction to the chapter

We first work only on $\mathbb{R}$.

Concrete question

Can one assign to every subset of $\mathbb{R}$ a number, its measure, which generalizes the length and remains additive?

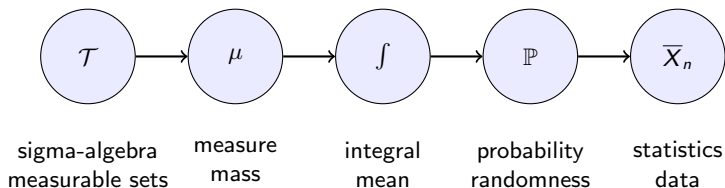
Two messages of the chapter

- Lebesgue outer measure gives a size to all subsets of $\mathbb{R}$.
- but it behaves as a genuine additive measure only on the measurable sets.

Motivation of the chapter 2

This difficulty will motivate the general notion of measurable space.

The five building blocks of the booklet



Language with probability and sets

Probability	sets
Sample space Ω	set Ω
Event A	Subset $A \subset \Omega$
Complementary event of A	Complement $\bar{A}$ or A^c
Event impossible	empty set $\emptyset$
Certain event	set Ω
A and B disjoint	A and B disjoint
Elementary event	singleton
Complete system of events	Partition of Ω

Partition and complete system of events

Definition

A **partition** of Ω is a family $(A_i)_{i \in I}$ of subsets of Ω such that:

- the A_i are pairwise disjoint;
- $\bigcup_{i \in I} A_i = \Omega$.

Probabilistic translation

A complete system of events is a partition of the sample space: each outcome belongs to exactly one part.

Definition (countable set)

Let E be a set. We say that E is **countable** if there exists an injective map

$$\varphi : E \longrightarrow \mathbb{N}.$$

Examples

$\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{N} \times \mathbb{N}$ are countable.

Non-examples

$\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$ are uncountable.

Warning: no uncountable sum

Example to remember

$$[0, 1] = \bigcup_{x \in [0, 1]} \{x\}, \quad \lambda^*([0, 1]) = 1, \quad \lambda^*({x}) = 0.$$

We cannot rigorously write

$$\lambda^*([0, 1]) = \sum_{x \in [0, 1]} \lambda^*({x}).$$

The expression $\sum_{x \in [0, 1]} 0$ is not a series: the index set $[0, 1]$ is uncountable.

Message

Additivity and subadditivity in measure theory apply to finite or countable families, not to arbitrary families.

Cantor numbering of $\mathbb{N} \times \mathbb{N}$

Explicit coding

$$\varphi(p, q) = \frac{(p + q)(p + q + 1)}{2} + q.$$

We scan $\mathbb{N} \times \mathbb{N}$ diagonal by diagonal.

	14	19	25	32	40
	9	13	18	24	31
	5	8	12	17	23
	2	4	7	11	16
q	0	1	3	6	10
	p				

Method: showing that a set is countable

Two frequent strategies

- 1 Directly exhibit an injection $\varphi : E \rightarrow \mathbb{N}$, or a bijection if we want to show that E is countably infinite.
- 2 Write E as a countable union of countable sets, or as the image of a countable set.

To show that a set is not countable

Cantor's diagonal argument is the standard tool, in particular for $\mathcal{P}(\mathbb{N})$ and $\mathbb{R}$.

Limit superior of a sequence of sets

Definition

Let $(A_n)_{n \in \mathbb{N}}$ be a sequence of subsets of Ω . The **limit superior** is the set of elements which belong to infinitely many of A_n :

$$\limsup_{n \in \mathbb{N}} A_n = \{\omega \in \Omega \mid \omega \in A_n \text{ for infinitely many indices } n\}.$$

Quantified formulation

$$\omega \in \limsup_n A_n \iff \forall i \in \mathbb{N}, \exists j \geq i, \omega \in A_j.$$

Limit inferior of a sequence of sets

Definition

The **limit inferior** is the set of elements which belong to all the A_n , except possibly for finitely many of them:

$$\liminf_{n \in \mathbb{N}} A_n = \{\omega \in \Omega \mid \omega \in A_n \text{ for every } n \text{ large enough}\}.$$

Quantified formulation

$$\omega \in \liminf_n A_n \iff \exists i \in \mathbb{N}, \forall j \geq i, \omega \in A_j.$$

Example: repeated coin tosses

We toss a coin infinitely many times:

$$\Omega = \{P, F\}^{\mathbb{N}}.$$

For $i \geq 1$, we set

$$A_i = \{\omega \in \Omega \mid \omega_i = F, \omega_{i+1} = F\}.$$

Asymptotic event

The event “the coin lands tails twice in a row infinitely many times” is

$$\limsup_{n \geq 1} A_n.$$

Proposition

Let $(A_n)_{n \in \mathbb{N}}$ be a sequence of subsets of Ω . Then

$$\limsup_{n \in \mathbb{N}} A_n = \bigcap_{n \in \mathbb{N}} \left(\bigcup_{m \geq n} A_m \right),$$

$$\liminf_{n \in \mathbb{N}} A_n = \bigcup_{n \in \mathbb{N}} \left(\bigcap_{m \geq n} A_m \right).$$

Proof: formula for the limit superior

Proof

For $\omega \in \Omega$:

$$\omega \in \bigcap_{n \in \mathbb{N}} \left(\bigcup_{m \geq n} A_m \right)$$

is equivalent to

$$\forall n \in \mathbb{N}, \omega \in \bigcup_{m \geq n} A_m,$$

therefore to

$$\forall n \in \mathbb{N}, \exists m \geq n, \omega \in A_m.$$

This means exactly that ω belongs to infinitely many of A_m , that is

$$\omega \in \limsup_n A_n.$$

Proof: formula for the limit inferior

Proof

For $\omega \in \Omega$:

$$\omega \in \bigcup_{n \in \mathbb{N}} \left(\bigcap_{m \geq n} A_m \right)$$

is equivalent to

$$\exists n \in \mathbb{N}, \omega \in \bigcap_{m \geq n} A_m,$$

therefore to

$$\exists n \in \mathbb{N}, \forall m \geq n, \omega \in A_m.$$

This means that ω belongs to all the A_m from some index onward:

$$\omega \in \liminf_n A_n.$$

Exercise: passing to complements

Statement

Show that

$$\overline{\limsup_n A_n} = \liminf_n \overline{A_n}, \quad \overline{\liminf_n A_n} = \limsup_n \overline{A_n}.$$

Idea of the solution

We start from the formulas with unions/intersections and apply De Morgan's laws:

$$\overline{\bigcap_n \bigcup_{m \geq n} A_m} = \bigcup_n \bigcap_{m \geq n} \overline{A_m}.$$

Method: handling the lim sup and lim inf

- 1 Translate the statement into words: “infinitely many times” or “from some index onward”.
- 2 Rewrite with the formulas

$$\limsup A_n = \bigcap_n \bigcup_{m \geq n} A_m, \quad \liminf A_n = \bigcup_n \bigcap_{m \geq n} A_m.$$

- 3 Use De Morgan for the complements.
- 4 In a probabilistic context, think about asymptotic events.

Probability on a finite sample space

Simple case

If Ω is finite, we can often define a probability on all subsets of Ω :

$$\mathcal{P}(\Omega) = \{A \mid A \subset \Omega\}.$$

Rolling a fair die

$$\Omega = \{1, 2, 3, 4, 5, 6\}, \quad \mathbb{P}(A) = \frac{|A|}{6}.$$

For example $\mathbb{P}(\{2, 4, 6\}) = \frac{3}{6} = \frac{1}{2}$.

Basic formulas

For every probability, we also want to keep the identities

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Can one do the same on an infinite sample space?

question

Can one define a natural probability on all subsets of $[0, 1]$, that is on $\mathcal{P}([0, 1])$?

Expected formulas

If A and B are disjoint,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

We also want

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

In general,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B),$$

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B).$$

The problem of measure

Starting question

Can one assign to all parts $A \subset \mathbb{R}$ a number $\lambda(A)$ representing its length?

Desired properties

At a minimum, we would like:

- $\lambda([a, b]) = b - a$;
- $\lambda(A + x) = \lambda(A)$;
- if $A_1, A_2, \dots$ are disjoint, then

$$\lambda\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n \geq 1} \lambda(A_n).$$

Answer

No, not on all the subsets of $\mathbb{R}$.

Definition: outer Lebesgue measure

Definition

For all parts $A \subset \mathbb{R}$, we define

$$\lambda^*(A) = \inf \left\{ \sum_{n \geq 1} (b_n - a_n) \mid A \subset \bigcup_{n \geq 1}]a_n, b_n[\right\}.$$

Interpretation

We measure A of the outer: we cover it by open intervals and we look for the minimal cost.

Covers: cost of a cover



If $A \subset \bigcup_n I_n$

The cost of the cover is

$$\sum_{n \geq 1} |I_n|,$$

and the outer measure therefore satisfies

$$\lambda^*(A) \leq \sum_{n \geq 1} |I_n|.$$

First properties of λ^*

Proposition

For all parts $A, B, A_1, A_2, \dots$ of $\mathbb{R}$:

- 1 $\lambda^*(\emptyset) = 0$;
- 2 if $A \subset B$, then $\lambda^*(A) \leq \lambda^*(B)$;
- 3 $\lambda^*\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n \geq 1} \lambda^*(A_n)$.

Important

The third property is an inequality: λ^* is not yet a true measure.

Proof of properties immediate (1/2)

Proof: empty set and monotonicity

- ① Let $n \in \mathbb{N}$. As

$$\emptyset \subset]0, 2^{-n}[,$$

we have, by definition of the infimum,

$$0 \leq \lambda^*(\emptyset) \leq 2^{-n}.$$

Letting n to $+\infty$, we get $\lambda^*(\emptyset) = 0$.

- ② Assume $A \subset B$. Let $(I_k)_{k \geq 1}$ be a cover of B by open intervals. Then

$$A \subset B \subset \bigcup_{k \geq 1} I_k.$$

The same cover is therefore admissible for A . Taking the infimum over all the covers of B , we obtain

$$\lambda^*(A) \leq \lambda^*(B).$$

Proof of properties immediate (2/3)

Proof: choice of covers

Let $\varepsilon > 0$. For each $n \geq 1$, by definition of the infimum, we choose a cover

$$A_n \subset \bigcup_{k \geq 1} I_{n,k}$$

such that

$$\sum_{k \geq 1} |I_{n,k}| \leq \lambda^*(A_n) + \varepsilon 2^{-n}.$$

Then the family of all the intervals $I_{n,k}$ covers $\bigcup_{n \geq 1} A_n$, therefore

$$\lambda^* \left(\bigcup_{n \geq 1} A_n \right) \leq \sum_{n \geq 1} \sum_{k \geq 1} |I_{n,k}|.$$

Proof of properties immediate (3/3)

Proof: detail of the inequality finale

For every n , we have

$$\sum_{k \geq 1} |I_{n,k}| \leq \lambda^*(A_n) + \varepsilon 2^{-n}.$$

Summing these inequalities over $n \geq 1$, we get

$$\sum_{n \geq 1} \sum_{k \geq 1} |I_{n,k}| \leq \sum_{n \geq 1} (\lambda^*(A_n) + \varepsilon 2^{-n}).$$

Therefore

$$\sum_{n \geq 1} \sum_{k \geq 1} |I_{n,k}| \leq \sum_{n \geq 1} \lambda^*(A_n) + \varepsilon \sum_{n \geq 1} 2^{-n} = \sum_{n \geq 1} \lambda^*(A_n) + \varepsilon.$$

Thus $\lambda^*(\cup_n A_n) \leq \sum_n \lambda^*(A_n) + \varepsilon$. Let ε to 0.

Proposition

For all parts $A \subset \mathbb{R}$:

- ① for every $x \in \mathbb{R}$, $\lambda^*(A + x) = \lambda^*(A)$;
- ② for every $\alpha \in \mathbb{R}$, $\lambda^*(\alpha A) = |\alpha| \lambda^*(A)$.

Idea

A cover of A by intervals is Mellin transform into a cover of $A + x$ by translation, or of αA by homothety. The lengths are preserved or multiplied by $|\alpha|$.

Proof of the invariance

Proof

If $A \subset \bigcup_n]a_n, b_n[$, then

$$A + x \subset \bigcup_n]a_n + x, b_n + x[$$

and the lengths are unchanged. Therefore $\lambda^*(A + x) \leq \lambda^*(A)$. Applying this inequality to $A + x$ and to the translation $-x$, we obtain the reverse inequality. For the homothety, if $\alpha > 0$,

$$\alpha A \subset \bigcup_n]\alpha a_n, \alpha b_n[$$

and the cost is multiplied by α . If $\alpha < 0$, we swap the bounds. If $\alpha = 0$, αA is empty or a singleton, therefore outer measure zero.

Outer measure intervals

Goal

We want to check that the outer measure extends the usual length:

$$\lambda^*([a, b[) = \lambda^*([a, b]) = b - a.$$

then, half-open intervals follow by encadrement.

Example: measure of an open interval (1/4)

Proposition

If $a < b$, then

$$\lambda^*(]a, b[) = b - a.$$

first inequality

As $]a, b[$ is an open interval and

$$]a, b[\subset]a, b[,$$

the cover consisting of the single interval $]a, b[$ is admissible. Its cost is $b - a$. Therefore

$$\lambda^*(]a, b[) \leq b - a.$$

Example: measure of an open interval (2/4)

Reminder: compactness

From every open cover of a compact set, we can extract a finite subcover. In other words, if

$$K \subset \bigcup_{n \geq 1} O_n,$$

then there exist $n_1, \dots, n_N$ such that

$$K \subset \bigcup_{j=1}^N O_{n_j}.$$

Example: measure of an open interval (3/4)

second inequality: reduction to a compact

Let $(I_n)_{n \geq 1}$ be a cover arbitrary of $]a, b[$ by open intervals. Fix $\eta > 0$ such that $2\eta < b - a$. Then

$$[a + \eta, b - \eta] \subset]a, b[\subset \bigcup_{n \geq 1} I_n.$$

By compactness of $[a + \eta, b - \eta]$, there exist $n_1, \dots, n_N$ such that

$$[a + \eta, b - \eta] \subset \bigcup_{j=1}^N I_{n_j}.$$

An interval compact of length $b - a - 2\eta$ cannot be covered by a family finite intervals including the sum of lengths is smallest that its length. Therefore

$$b - a - 2\eta \leq \sum_{j=1}^N |I_{n_j}|.$$

Example: measure of an open interval (4/4)

second inequality: passage to the infimum

As $\{n_1, \dots, n_N\}$ is an under-family of integers, we have

$$\sum_{j=1}^N |I_{n_j}| \leq \sum_{n \geq 1} |I_n|.$$

Therefore, for every cover (I_n) of $]a, b[$,

$$b - a - 2\eta \leq \sum_{n \geq 1} |I_n|.$$

Taking the infimum over all the admissible covers,

$$b - a - 2\eta \leq \lambda^*(]a, b[).$$

then, letting η to 0,

$$b - a \leq \lambda^*(]a, b[).$$

With the first inequality: $\lambda^*(]a, b[) = b - a$.

Example: measure of a closed interval

Proposition

If $a < b$, then

$$\lambda^*([a, b]) = b - a.$$

Proof

For every $\varepsilon > 0$,

$$[a, b] \subset]a - \varepsilon, b + \varepsilon[, \quad |]a - \varepsilon, b + \varepsilon[| = b - a + 2\varepsilon.$$

Therefore $\lambda^*([a, b]) \leq b - a + 2\varepsilon$ for every $\varepsilon > 0$, then $\lambda^*([a, b]) \leq b - a$. On the other hand, $]a, b[\subset [a, b]$, therefore by monotonicity

$$b - a = \lambda^*([a, b]) \geq \lambda^*([a, b]) = b - a.$$

The two inequalities give $\lambda^*([a, b]) = b - a$.

Theorem: outer measure intervals

Theorem

For $a < b$, we have

$$\lambda^*([a, b[) = \lambda^*([a, b]) = \lambda^*([a, b[) = \lambda^*([a, b]) = b - a.$$

Proof

We comes to prove the two extreme cases:

$$\lambda^*([a, b[) = b - a, \quad \lambda^*([a, b]) = b - a.$$

Now

$$]a, b[\subset [a, b[\subset [a, b], \quad]a, b[\subset]a, b] \subset [a, b].$$

By monotonicity and squeezing, the two half-open intervals also have outer measure $b - a$.

Example: measure of $\mathbb{R}$

Proposition

$$\lambda^*(\mathbb{R}) = +\infty.$$

Proof

For every $M > 0$, we have $[-M, M] \subset \mathbb{R}$, therefore by monotonicity

$$2M = \lambda^*([-M, M]) \leq \lambda^*(\mathbb{R}).$$

As M is arbitrary, $\lambda^*(\mathbb{R})$ cannot be finite.

Example: measure of a singleton

Proposition

For every $x \in \mathbb{R}$,

$$\lambda^*(\{x\}) = 0.$$

Proof

For every $\varepsilon > 0$,

$$\{x\} \subset]x - \varepsilon, x + \varepsilon[, \quad |]x - \varepsilon, x + \varepsilon|| = 2\varepsilon.$$

Therefore $0 \leq \lambda^*(\{x\}) \leq 2\varepsilon$ for every $\varepsilon > 0$, hence $\lambda^*(\{x\}) = 0$.

The outer measure of $\mathbb{Q}$ is zero

Example

We enumerate $\mathbb{Q} = \{q_0, q_1, q_2, \dots\}$. For every $\varepsilon > 0$, we choose intervals centered in q_i of lengths

$$\frac{\varepsilon}{2}, \quad \frac{\varepsilon}{4}, \quad \frac{\varepsilon}{8}, \quad \frac{\varepsilon}{16}, \quad \dots$$

They cover $\mathbb{Q}$ and their total cost is

$$\frac{\varepsilon}{2} + \frac{\varepsilon}{4} + \frac{\varepsilon}{8} + \frac{\varepsilon}{16} + \dots = \varepsilon.$$

Conclusion

$$\lambda^*(\mathbb{Q}) = 0.$$

More generally, every countable subset of $\mathbb{R}$ has outer measure zero.

Definition

In a measure space $(\Omega, \mathcal{T}, \mu)$, a measurable set $N \in \mathcal{T}$ is **null** if

$$\mu(N) = 0.$$

For Lebesgue measure

Singletons, $\mathbb{Q}$ and more generally all countable subsets of $\mathbb{R}$ are null.

Why is λ^* not enough?

Problem

A measure must be additive on countable disjoint unions. The outer measure λ^* is only subadditive on all subsets of $\mathbb{R}$.

Idea of Carathéodory

We keep the sets B which split every set A cleanly:

$$\lambda^*(A) = \lambda^*(A \cap B) + \lambda^*(A \cap B^c).$$

Definition

A part $B \subset \mathbb{R}$ is called **Lebesgue-measurable** if

$$\forall A \subset \mathbb{R}, \quad \lambda^*(A) = \lambda^*(A \cap B) + \lambda^*(A \cap B^c).$$

We denote by $\mathcal{M}(\lambda^*)$ the set of Lebesgue-measurable sets of $\mathbb{R}$.

Geometric reading

B is measurable if the decomposition by B and B^c neither loses nor creates measure.

Why introduce a sigma-algebra?

Probability formulas impose stability properties

To compute with events, we want to be able to form and measure

$$\emptyset, \quad A^c, \quad A \cup B, \quad A \cap B.$$

The usual formulas therefore require these sets to remain measurable:

$$\mathbb{P}(\emptyset) = 0, \quad \mathbb{P}(A^c) = 1 - \mathbb{P}(A),$$

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Idea

A sigma-algebra is precisely a family of sets stable enough to perform these calculations: it is exactly what is needed for $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$, $\mathbb{P}(A^c)$ and $\mathbb{P}(\emptyset)$ to make sense.

Reminder: what is a sigma-algebra?

Definition

A sigma-algebra on a set Ω is a set $\mathcal{T} \subset \mathcal{P}(\Omega)$ such that:

- ① $\Omega \in \mathcal{T}$;
- ② if $A \in \mathcal{T}$, then $A^c \in \mathcal{T}$;
- ③ if $(A_n)_{n \geq 1}$ is a sequence elements of $\mathcal{T}$, then $\bigcup_{n \geq 1} A_n \in \mathcal{T}$.

Two extreme examples

The trivial sigma-algebra is $\{\emptyset, \Omega\}$, and the discrete sigma-algebra is $\mathcal{P}(\Omega)$.

Theorem

The set $\mathcal{M}(\lambda^*)$ of Lebesgue-measurable sets satisfies:

- 1 $\emptyset \in \mathcal{M}(\lambda^*)$;
- 2 if $B \in \mathcal{M}(\lambda^*)$, then $B^c \in \mathcal{M}(\lambda^*)$;
- 3 if $(B_n)_{n \geq 1}$ is a sequence elements of $\mathcal{M}(\lambda^*)$, then $\bigcup_{n \geq 1} B_n \in \mathcal{M}(\lambda^*)$;
- 4 every subset of outer measure zero is in $\mathcal{M}(\lambda^*)$.

Moreover, λ^* is additive on countable disjoint unions of measurable sets.

Proof of Carathéodory: empty set

Proof of Point 1

For every $A \subset \mathbb{R}$, we have

$$A \cap \emptyset = \emptyset, \quad A \cap \emptyset^c = A \cap \mathbb{R} = A.$$

As $\lambda^*(\emptyset) = 0$, we get

$$\lambda^*(A \cap \emptyset) + \lambda^*(A \cap \emptyset^c) = \lambda^*(\emptyset) + \lambda^*(A) = 0 + \lambda^*(A) = \lambda^*(A).$$

Therefore $\emptyset$ satisfies the Carathéodory criterion: $\emptyset \in \mathcal{M}(\lambda^*)$.

Proof of Carathéodory: complement

Proof of Point 2

Let $B \in \mathcal{M}(\lambda^*)$. Then, for every $A \subset \mathbb{R}$,

$$\lambda^*(A) = \lambda^*(A \cap B) + \lambda^*(A \cap B^c).$$

Now

$$A \cap (B^c)^c = A \cap B.$$

We can therefore rewrite the previous equality in the form

$$\lambda^*(A) = \lambda^*(A \cap B^c) + \lambda^*(A \cap (B^c)^c).$$

It is exactly the Carathéodory criterion for B^c . Therefore $B^c \in \mathcal{M}(\lambda^*)$.

Proof of Carathéodory: finite union

Preparatory step

Let $B_1, B_2 \in \mathcal{M}(\lambda^*)$. For every $A \subset \mathbb{R}$, by first cutting with B_1 :

$$\lambda^*(A) = \lambda^*(A \cap B_1) + \lambda^*(A \cap B_1^c).$$

then by cutting $A \cap B_1^c$ with B_2 :

$$\lambda^*(A \cap B_1^c) = \lambda^*(A \cap B_1^c \cap B_2) + \lambda^*(A \cap B_1^c \cap B_2^c).$$

Therefore

$$\lambda^*(A) = \lambda^*(A \cap B_1) + \lambda^*(A \cap B_1^c \cap B_2) + \lambda^*(A \cap B_1^c \cap B_2^c).$$

Proof of Carathéodory: finite union

Conclusion of the preparatory step

We have

$$A \cap (B_1 \cup B_2) = (A \cap B_1) \sqcup (A \cap B_1^c \cap B_2),$$

and

$$A \cap (B_1 \cup B_2)^c = A \cap B_1^c \cap B_2^c.$$

The subadditivity gives

$$\lambda^*(A \cap (B_1 \cup B_2)) \leq \lambda^*(A \cap B_1) + \lambda^*(A \cap B_1^c \cap B_2).$$

Adding $\lambda^*(A \cap (B_1 \cup B_2)^c)$ and using the equality of the previous slide:

$$\lambda^*(A \cap (B_1 \cup B_2)) + \lambda^*(A \cap (B_1 \cup B_2)^c) \leq \lambda^*(A).$$

The other inequality comes from subadditivity applied to the decomposition of A . Therefore $B_1 \cup B_2$ is measurable.

Proof of Carathéodory: a countable union

Proof of Point 3, disjoint case

Let $(B_k)_{k \geq 0}$ be a pairwise disjoint family of measurable sets. By decomposing successively with $B_0, \dots, B_m$, we obtain by induction

$$\lambda^*(A) = \sum_{k=0}^m \lambda^*(A \cap B_k) + \lambda^*\left(A \cap \bigcap_{k=0}^m B_k^c\right).$$

As

$$A \cap \bigcap_{k \geq 0} B_k^c \subset A \cap \bigcap_{k=0}^m B_k^c,$$

monotonicity gives

$$\lambda^*(A) \geq \sum_{k=0}^m \lambda^*(A \cap B_k) + \lambda^*\left(A \cap \bigcap_{k \geq 0} B_k^c\right).$$

We then let m tend to $+\infty$.

Proof of Carathéodory: a countable union

Details of the second inequality

Set $B = \bigcup_{k \geq 0} B_k$. The previous slide gives

$$\lambda^*(A) \geq \sum_{k \geq 0} \lambda^*(A \cap B_k) + \lambda^*(A \cap B^c).$$

On the other hand, as the B_k are disjoint,

$$A \cap B = \bigcup_{k \geq 0} (A \cap B_k).$$

By subadditivity,

$$\lambda^*(A \cap B) \leq \sum_{k \geq 0} \lambda^*(A \cap B_k).$$

Therefore

$$\lambda^*(A) \geq \lambda^*(A \cap B) + \lambda^*(A \cap B^c).$$

The reverse inequality is the subadditivity applied to $A = (A \cap B) \cup (A \cap B^c)$.
Therefore B is measurable.

Proof of Carathéodory: arbitrary union

End of the proof of Point 3

If the B_k are not disjoint, we replace them with a disjoint family:

$$C_0 = B_0, \quad C_k = B_k \cap \left(\bigcup_{j < k} B_j \right)^c \quad (k \geq 1).$$

By the points already proved, the C_k are measurable: we use finite unions, complements and finite intersections. They are pairwise disjoint and

$$\bigcup_{k \geq 0} C_k = \bigcup_{k \geq 0} B_k.$$

The disjoint case therefore shows that $\bigcup_k B_k$ is measurable. Thus $\mathcal{M}(\lambda^*)$ is indeed a sigma-algebra.

Proof of Carathéodory: measure zero

Proof of Point 4

Let $B \subset \mathbb{R}$ such that $\lambda^*(B) = 0$. For every $A \subset \mathbb{R}$, by monotonicity,

$$0 \leq \lambda^*(A \cap B) \leq \lambda^*(B) = 0,$$

therefore $\lambda^*(A \cap B) = 0$. By subadditivity,

$$\lambda^*(A) \leq \lambda^*(A \cap B) + \lambda^*(A \cap B^c) = 0 + \lambda^*(A \cap B^c).$$

By monotonicity, as $A \cap B^c \subset A$,

$$\lambda^*(A \cap B^c) \leq \lambda^*(A).$$

The two inequalities give the Carathéodory criterion for B .

Proof of Carathéodory: additivity

Proof

Let (B_k) be a pairwise disjoint family of measurable sets and set $B = \bigcup_k B_k$. The proof of the disjoint case gives, for every m ,

$$\lambda^*(B) \geq \sum_{k=0}^m \lambda^*(B_k).$$

Letting m to $+\infty$,

$$\lambda^*(B) \geq \sum_{k \geq 0} \lambda^*(B_k).$$

The reverse inequality is the countable subadditivity:

$$\lambda^*(B) = \lambda^*\left(\bigcup_k B_k\right) \leq \sum_{k \geq 0} \lambda^*(B_k).$$

Therefore $\lambda^*(\bigcup_k B_k) = \sum_k \lambda^*(B_k)$.

Theorem of Vitali

Theorem

[Vitali set] There exists a part F of $[0, 1]$ which is not Lebesgue-measurable:

$$F \notin \mathcal{M}(\lambda^*).$$

In particular,

$$\mathcal{M}(\lambda^*) \subsetneq \mathcal{P}(\mathbb{R}).$$

Message

We cannot measure all the subsets of $\mathbb{R}$ in conservant the natural properties of the length.

Construction of the Vitali set

We $\mathbb{R}$, we define the equivalence relation

$$x \sim y \iff x - y \in \mathbb{Q}.$$

Choice of a representing

We choose a set $F \subset [0, 1]$ containing exactly a representing of each equivalence class which rencontre $[0, 1]$.

Family of translates

We set

$$Q = \mathbb{Q} \cap [-1, 1]$$

and we consider the translates

$$(q + F)_{q \in Q}.$$

The translates are pairwise disjoint

Proof

Assume that

$$(q_1 + F) \cap (q_2 + F) \neq \emptyset$$

with $q_1 \neq q_2$. Then there exist $x, y \in F$ such that

$$q_1 + x = q_2 + y.$$

Therefore

$$x - y = q_2 - q_1 \in \mathbb{Q} \setminus \{0\}.$$

Thus $x \sim y$ with $x \neq y$, which contradicts the fact that F contains a unique representative in each equivalence class.

Bounds for the translates

Inclusion upper

As $F \subset [0, 1]$ and $Q \subset [-1, 1]$,

$$\bigcup_{q \in Q} (q + F) \subset [-1, 2].$$

Inclusion lower

every $x \in [0, 1]$ is equivalent to a unique $y \in F$, therefore $x - y \in \mathbb{Q}$. As $x, y \in [0, 1]$,

$$x - y \in [-1, 1] \cap \mathbb{Q} = Q.$$

Therefore

$$[0, 1] \subset \bigcup_{q \in Q} (q + F).$$

Contradiction if F was measurable

Proof

Assume F measurable. The translates $q + F$ are measurable, pairwise disjoint, and all have the same outer measure as F . Therefore

$$\lambda^* \left(\bigcup_{q \in Q} (q + F) \right) = \sum_{q \in Q} \lambda^*(F).$$

By the previous squeeze estimate,

$$1 = \lambda^*([0, 1]) \leq \sum_{q \in Q} \lambda^*(F) \leq \lambda^*([-1, 2]) = 3.$$

but Q is countable infinite: if $\lambda^*(F) = 0$, the sum is 0; if $\lambda^*(F) > 0$, it is $+\infty$. In the two cases, contradiction.

Conclusion of the chapter 1

The problem of measure has no solution on $\mathcal{P}(\mathbb{R})$

No measure that is additive over countable disjoint unions and has value $b - a$ on intervals $]a, b[$ can be defined on all of $\mathcal{P}(\mathbb{R})$.

This that we must sacrifice

One must necessarily abandonner one of exigences following:

- measure all the subsets of $\mathbb{R}$;
- keep additivity;
- keep the natural properties of length.

Choice of Lebesgue

We measure the sets belonging to a large sigma-algebra of well-behaved subsets: the Lebesgue-measurable sets.

What to know after CM1

- 1 Translate the probabilistic operations into set operations.
- 2 Handle countability, $\limsup$ and $\liminf$ sets.
- 3 Use the definition of λ^* with covers.
- 4 Prove that a countable set is of outer measure zero.
- 5 Use the Carathéodory criterion in simple cases.
- 6 Explain the idea of the Vitali set.